Solutions Tutorial 3

Exercise 1

(a) Apply the expected utility representation to each reported indifference. Indifference means the two lotteries have the same expected utility.

For €20: $u(20) = 0.6\,u(0) + 0.4\,u(100) = 0.6 \cdot 0 + 0.4 \cdot 1 = 0.4$.

For €50: $u(50) = 0.3\,u(0) + 0.7\,u(100) = 0.7$.

This is step 1 of the proof of the expected utility theorem (L7, appendix) carried out with numbers: each sure outcome is calibrated against a mixture of the best and worst outcomes, and the calibration weight *is* its utility.

(b) Expected utilities and expected values:

	expected utility	expected value (€)
A	$0.3(0.4) + 0.3(0.7) + 0.3(1) = 0.63$	$0.3(20) + 0.3(50) + 0.3(100) = 51$
B	0.7	50
C	0.65	65

The subject ranks $B \succ C \succ A$. By expected value the ranking is $C > A > B$: the sure €50 is the *lowest*-mean option and yet the most preferred, and C has a mean €15 higher than B but is worse for this subject. This is risk aversion at work.

(c) The lottery has expected utility $\frac{1}{2}(0.4) + \frac{1}{2}(1) = 0.7 = u(50)$. Since u is strictly increasing, the certainty equivalent is unique: $c =$ €50. The expected value is €60, so the risk premium is $60 - 50 =$ €10.

Exercise 2

(a) Without insurance, income is 100 with probability 0.95 and 40 with probability 0.05.

- Mean: $0.95(100) + 0.05(40) = 97$.
- Expected utility: $0.95 \ln 100 + 0.05 \ln 40 = 0.95(4.6052) + 0.05(3.6889) = 4.5594$.
- Certainty equivalent: $c = e^{4.5594} = 95.52$.
- Risk premium: $97 - 95.52 = 1.48$.

Under full coverage their income is $w - P$ in both states, where P is the total price. The most they would pay solves $\ln(100 - P) = 4.5594$, i.e. $P = 100 - 95.52 = 4.48$. This is the expected loss plus the risk premium: $\pi D + 1.48 = 3 + 1.48$. The fair price is $\pi D = 3$, and the price they face is $qD = 3.6$. Since $3.6 < 4.48$, they strictly prefer full coverage to none. Part (c) shows they do better still with partial coverage.

(b) Concavity. $c_1(a)$ and $c_2(a)$ are affine in a, and u is concave, so $u(c_1(a))$ and $u(c_2(a))$ are concave in a (a concave function of an affine function is concave). V is a sum of these with positive weights, so V is concave.

Derivative. Since $c_1'(a) = -q$ and $c_2'(a) = 1 - q$,

$$V'(a) = -q(1-\pi)\,u'(c_1(a)) + (1-q)\pi\,u'(c_2(a)).$$

Evaluate at full coverage. At $a = D$, $c_1(D) = w - qD = c_2(D)$: consumption is equal across states. So $u'(c_1(D)) = u'(c_2(D)) = u'(w - qD) > 0$, and

$$V'(D) = u'(w-qD)\,\big[\pi(1-q) - q(1-\pi)\big] = u'(w-qD)\,(\pi - q).$$

Conclude. By the fact given in the stem, $a = D$ is optimal iff $V'(D) \geq 0$, i.e. iff $\pi \geq q$.

When insurance is actuarially fair, $q = \pi$, so full coverage is optimal for *every* risk-averse expected utility maximiser, whatever the curvature of u. With any loading ($q > \pi$), $V'(D) < 0$ and they buy strictly less than full coverage.

(c) FOC. With $u = \ln$, V is strictly concave, so an interior solution is characterised by $V'(a) = 0$:

$$\frac{q(1-\pi)}{w - qa} = \frac{\pi(1-q)}{w - D + (1-q)a}.$$

Solve. Cross-multiply:

$$q(1-\pi)(w-D) + q(1-q)(1-\pi)\,a = \pi(1-q)\,w - \pi(1-q)q\,a.$$

Collect the terms in a; their coefficient is $q(1-q)[(1-\pi)+\pi] = q(1-q)$:

$$q(1-q)\,a = \pi(1-q)\,w - q(1-\pi)(w-D) \quad\Longrightarrow\quad a^* = \frac{\pi w}{q} - \frac{(1-\pi)(w-D)}{1-q}.$$

Evaluate. $a^* = 0.05(100)/0.06 - 0.95(40)/0.94 = 83.33 - 40.43 = 42.91$. This lies in $(0, 60)$, so it is indeed the optimum. (Check: with $q = \pi$ the formula gives $w - (w - D) = D$, matching (b).)

Certainty equivalents.

coverage	c_1	c_2	certainty equivalent
$a = 0$	100	40	95.52
$a^* = 42.9$	97.43	80.33	96.49
$a = D = 60$	96.40	96.40	96.40

Insurance is worth about 1 unit of sure income to the worker. Moving from full to optimal coverage gains only 0.09: the loss from bearing a little risk is second order, while the saving on the loading is first order.

Wealth. Differentiating the formula,

$$\frac{\partial a^*}{\partial w} = \frac{\pi}{q} - \frac{1-\pi}{1-q} = \frac{\pi(1-q) - q(1-\pi)}{q(1-q)} = \frac{\pi - q}{q(1-q)},$$

which is negative when $q > \pi$ (-0.18 here). Log utility has $r_A(x) = 1/x$, decreasing in x: richer workers are less averse to a given euro risk, so they buy less of an insurance product sold above its fair price. The flip side: at loaded prices, it is the poorer workers who value insurance most.

(d) With $u(x) = -e^{-\alpha x}$, $u'(x) = \alpha e^{-\alpha x}$. The FOC $V'(a) = 0$ becomes

$$q(1-\pi)\,\alpha e^{-\alpha c_1} = \pi(1-q)\,\alpha e^{-\alpha c_2} \iff e^{\alpha(c_2-c_1)} = \frac{\pi(1-q)}{q(1-\pi)} \iff c_1 - c_2 = \frac{1}{\alpha}\ln\left[\frac{q(1-\pi)}{\pi(1-q)}\right].$$

Since $c_1 - c_2 = (w - qa) - (w - D + (1-q)a) = D - a$,

$$a^* = D - \frac{1}{\alpha}\ln\left[\frac{q(1-\pi)}{\pi(1-q)}\right].$$

With $\alpha = 0.02$: $\frac{0.06 \cdot 0.95}{0.05 \cdot 0.94} = 1.2128$, $\ln 1.2128 = 0.193$, so $a^* = 60 - 9.64 = 50.36$.

w drops out because the FOC depends only on the *gap* $c_1 - c_2$: with $u = -e^{-\alpha x}$, adding a constant to both consumption levels multiplies both marginal utilities by the same factor. This is constant absolute risk aversion. Contrast (c): there, wealth mattered because r_A falls with wealth.